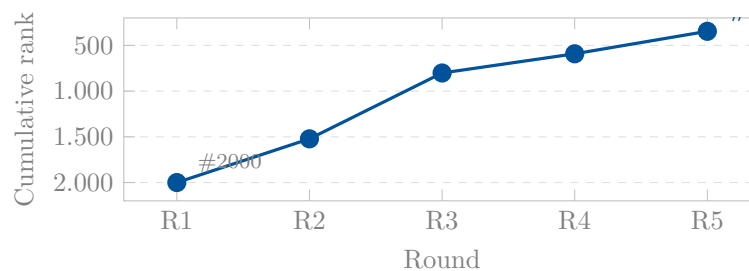

IMC Prosperity 4

Executive Extract

*Lessons from a solo entry in IMC's 18,803-team
algorithmic-trading and game-theory simulation*

Yasser EL KOUHEN | 9 May 2026



Cumulative overall-leaderboard rank, monotone improvement R1 → R5.

Why I entered

In late February I missed an in-person Natixis CIB recruiting event in London because the flights from Paris were too expensive that week. Three weeks later, I made up for it: I paid double the cost of the missed trip to attend the ETH Zurich Kontakt Party in mid-March. That evening I met quants and recruiters from Jane Street, Citadel, Hudson River Trading, Optiver, IMC Trading, and Flow Traders. By the end of it I had decided what I want to do: become a quantitative researcher / trader at a top market-maker.

I had been carrying the wrong picture of the field — I had assumed quant work was beyond my skill ceiling and culturally too far from where I wanted to be. Zurich corrected both. The intellectual challenge was exactly what I had been missing since my French preparatory classes (MPSI / MP*), where I last did ‘real’ mathematics. The competitive environment, the kind of people whose work I could admire, and the path to genuine financial independence — the kind that meant my father, who raised our family on a low-middle-income French salary while counting cents, would never have to do that again — all ran through this kind of role. But I had no quant project on my CV, no proof I belonged. When I saw the IMC Prosperity 4 announcement two weeks later, I knew it was my first concrete step.

Seventeen days, 18,803 teams, 117 countries. Solo entry, no team. I started ranked around #2,000 after Round 1 and finished at **#346** — top 1.84%; country rank #11 in France; cumulative XIREC 383,727. I am proud of the trajectory. I also know it is far from where I want to end up.

This extract distills the ~ 11,300-line full report into the parts that mattered: the result, the three load-bearing Round-5 insights (backtester $\neq$ competition; adverse selection scales with volume; no tradable lead-lag at 100 ms resolution), the v42 final submitted strategy with its JSON-verified PnL decomposition, and what I learned. It targets a reader who wants the substance and the human narrative without the chronological deep-dive in the full report.

Final result. Overall rank **#346 / 18,803** (top 1.84%); algorithmic **#537**; **manual #204**; country **#11** (France; 1,549 universities, 117 countries). Final cumulative score **383,727**. Manual contributed 59.4% of cumulative R3+R4+R5; algorithmic 40.6%.

What I built. Solo competitor. 42 versioned Round-5 traders, 76 research scripts, manual derivations for all 5 rounds, a GSD-managed planning trail with 16 numbered phases, a ~ 115 -page Obsidian wiki second-brain, and an $\sim 11,300$ -line technical report (this extract is its 8-page distillation).

Background. MPSI / MP* (intensive French preparatory-class maths and physics) $\rightarrow$ MEng Computer Science / Machine Learning $\rightarrow$ MSc Data Science & Business Analytics. Strong mathematical / ML background; the market-finance domain knowledge documented here was acquired *during* the competition itself.

The Competition

IMC Prosperity 4 is a multi-round algorithmic-trading and game-theory competition: each team submits a single-file Python algorithm (`trader.py`) that runs per-tick against a simulated limit-order-book exchange, plus per-round manual puzzles. Profit is measured in synthetic XIREC units. Five rounds over ~ 17 days (April 14 – April 30, 2026); only Rounds 3, 4, and 5 contribute to the final cumulative score.

Per-round summary

Round	Algo	Manual	Total	Theme
R1	98,172	71,500	169,672	ACO mean-reversion + IPR trend; Intarian Welcome auction
R2	91,529	153,345	244,874	MAF auction $b^* = 3,000$; Invest & Expand (16, 48, 36)
R3	40,800	75,238	116,038	HYDROGEL AR(1) MM; VEV options; Bio-Pods Bayesian-Nash
R4	57,048	57,516	114,564	Mark counterparty taxonomy; AETHER exotic-options portfolio
R5	57,911	95,214	153,125	50 products / ± 10 limit; Ashflow Alpha allocation
Cumulative R3+R4+R5			383,727	XIREC

What I did in each round

Round 1 (algo + manual). ACO (ASH_COATED_OSMIUM) had strong AR(1) mean-reversion ($\rho_1 = -0.495$); IPR (INTARIAN_PEPPEER_ROOT) had a strong upward intraday drift. Two structurally opposite strategies: ACO with anchor-blended fair value $FV_t = 0.85 \cdot EMA_t + 0.15 \cdot 10,000$ and bucketed-AR(1) reversion correction (passive market-making); IPR with greedy long accumulation up to the position cap (99.2% of the structural ceiling, no sells). Manual: *Intarian Welcome* clearing-price auction — closed-form on (B, Q) given the sequential clearing rule; submitted Dryland Flax 5k @ 29 + Ember Mushroom 35k @ 18 for \$71,500.

Round 2 (algo continuation + manual-heavy). No new tradable products; ACO/IPR continued. Strategic emphasis shifted to the manual challenges: *MAF* (one-shot sealed-bid auction for a +25% quote-volume bonus, asymmetric-regret framing favoured \$3,000, accepted) and *Invest & Expand* (3-pillar resource-allocation game, $\log \times \text{linear} \times \text{rank}$ EV system, FOC numerical solution $(x_R, x_S, x_V) \approx (16, 48, 36)$, submitted (15, 50, 35)). Manual delivered \$153,345 — the largest manual contribution of any round.

Round 3 (algo: options arrive). All previous products retired. New: HYDROGEL_PACK (mean-reverting underlying), VELVETFRUIT_EXTRACT (options underlying), and a 10-strike VEV voucher chain (strikes 4,000–6,500, European calls). HYDROGEL got bucketed AR(1) market-making ($\rho_{\text{small}} = 0.08$, $\rho_{\text{large}} = 0.42$, anchor weight 0.20); VEV options got per-strike implied-volatility-surface calibration ($\sigma_K \in [0.230, 0.249]$ across the smile, ATM trough at strike 5,400) with two-level quoting and IV-z-score-biased sizing. Delta hedging passive only. The first official R3 backtest came back near \$0 after I had been at a steady \$10K through R2 — this was the lowest moment of the competition; Lessons Learned addresses it. Manual: *Bio-Pods* sealed-bid auction, symmetric Bayesian-Nash equilibrium at $(b_1, b_2) = (755, 840)$, $f^* = 81.67/\text{CP}$. Submitted the robust (760, 855); realised \$75,238 at 92% of NE optimum across $N = 1,000$ counterparties.

Round 4 (algo: counterparty data). HYDROGEL/VEV continued. New information: every trade now lists the named-bot counterparty (Mark_01, 14, 22, 38, 49, 55, 67). Built a counterparty taxonomy: Mark_67 is a dip-buyer with informed directional bias (+1,510 net VEL); Mark_49 is a local-high seller with 500 ms cooldown signal; Mark_22 is a high-frequency

OTM-call short-seller (excluded from `mark_net` composite to avoid drowning the signal); pairs `Mark_14/38` are tick-alternating market-makers (net ≈ 0). Composite signal `mark67 - mark49` biased VEL aggressiveness. Manual: *AETHER_CRYSTAL* 12-instrument exotic-options portfolio (chooser $K = 50$, binary-put $K = 40$ with bull-spread cliff hedge, knock-out put $K = 45/B = 35$ with Merton barrier + Broadie–Glasserman discrete adjustment). Pricing edges +\$0.30, +\$0.232, +\$0.045 respectively. Pre-trade $\mathbb{E}[+\$175,200]$; realised \$57,516 (33%). Shortfall was a path-dependent regime feature (V-shaped underlying knocked the long-KO-put hedges out), not a model error — peer position data showed every quantitatively-prepared team ran a substantively similar portfolio.

Round 5 (50 products, structural break). All previous products retired. New: 50 products in 10 categories (PEBBLES, SNACKPACK, PANEL, MICROCHIP, UV_VISOR, OXYGEN_SHAKE, GALAXY_SOUNDS, SLEEP_POD, ROBOT, TRANSLATOR), hard ± 10 position limit per product. $AR(1) \approx +0.999$ across all 50 products — effectively unit-root prices, no exploitable mean-reversion at the tick level; the only signal is multi-day directional drift. The strategy went through 42 versioned iterations (v1 $\rightarrow$ v42); the next sections cover the three load-bearing insights and the final v42 design. Manual: *Ashflow Alpha* allocation on 9 archetype-classified goods, quadratic-fee structure ($fee_i = (p_i/100)^2 \cdot B$), ridge-regression closed-form $p_i^* = s_i/2$. Submitted with 4-of-6 archetype classifications correct; realised \$95,214 ($\sim 68\%$ of model PnL).

The deep technical content of the rest of this extract focuses on **Round 5** because that is where the load-bearing methodological insights of the competition crystallised. The full chronological narrative for every round and every product is in the main report.

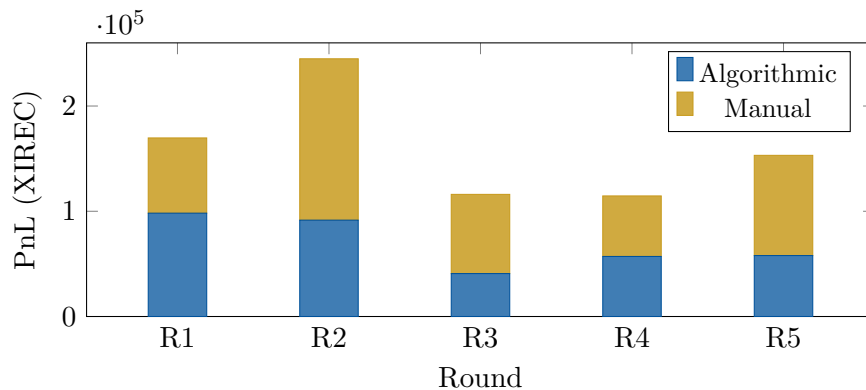


Figure 1: Per-round PnL by source. Manual game-theory puzzles delivered 59.4% of cumulative scoreative XIREC despite the algo representing the bulk of project hours.

The Three Round-5 Insights

Round 5 was the structural break of the competition: every prior product was retired, replaced by 50 new products in 10 categories with a hard ± 10 position limit. All 50 products had AR(1) coefficient $\approx +0.999$ (effectively unit-root); tradable mean-reversion disappeared, replaced by multi-day directional drift as the only extractable algorithmic alpha. The strategy went through 42 versioned iterations and surfaced three load-bearing insights:

Insight 1 — Backtester $\neq$ competition

I came to this insight at the worst possible time: with v36 sitting at the top of my local-backtester leaderboard and a submission deadline approaching. The numbers were so close that I almost just shipped the obvious winner.

The Prosperity local backtester runs only the *first* 10% of Day 4 ($\sim 100,000$ of 1,000,000 ticks). The competition scores the *full* Day 5. The two windows have different optimal strategies. For each version v with directional positions $q_p \in \{-10, 0, +10\}$ and per-product per-window mid-price drift $\Delta_p^W = \text{mid}_p[T_W] - \text{mid}_p[0]$, realised PnL is

$$\text{PnL}_p^W = q_p \Delta_p^W - \frac{1}{2} |q_p| s_p,$$

and the *conflict products* satisfy $\text{sign}(\Delta_p^{\text{bt}}) \neq \text{sign}(\Delta_p^{\text{full}})$. v36 was the backtester champion (\$78,799); the submitted **v42 deliberately scored \$72,000 — \$6,799 below v36** — to avoid optimising to the wrong window. On the full-day Day-5 score, this deliberate regression was the right call: estimated full-day gain +\$22,050 vs the backtester champion.

Insight 2 — Adverse selection scales linearly with volume

This one I learned by getting it wrong on purpose, in a sense. v40 was my hypothesis “if these TIER3 products lose at LIMIT 5, maybe they will recover at LIMIT 10 with more volume”. The hypothesis was wrong, and the doubling of the loss was the experimental proof of why.

For market-making at LIMIT ℓ on a tight-spread product with average captured spread $\bar{s}_p$ and per-fill adverse-selection cost α_p , expected PnL is

$$\mathbb{E}[\text{PnL}_p^{\text{MM}}(\ell)] \approx \kappa_p \ell (\bar{s}_p + \alpha_p),$$

linear in ℓ with sign determined by the bracket. The optimum is

$$\ell_p^* = \begin{cases} 10 & \bar{s}_p + \alpha_p > 0 \\ 5 \text{ (TIER3)} & \bar{s}_p + \alpha_p \approx 0 \\ 0 \text{ (BLACKLIST)} & \bar{s}_p + \alpha_p < 0. \end{cases}$$

v40 (not submitted) is the empirical demonstration: it removed TIER3 (LIMIT 5 $\rightarrow$ 10) on the false hypothesis “more trades = recovery”. The adverse-selection loss on the affected products *doubled* accordingly. v41 restored TIER3; v42 went further with 11 products on permanent LIMIT = 0 blacklist, 2 small-loss products at TIER3.

Insight 3 — No tradable lead-lag at 100 ms resolution

I went into this expecting to find something. The within-category contemporaneous correlations were striking ($\rho = -0.92$ on SNACKPACK chocolate / vanilla), and the natural follow-up was: do these moves happen simultaneously, or does one lead the other? A tradable lead

would have been the easiest alpha of Round 5. There was none.

A full lagged-CCF battery on $\binom{50}{2} = 1,225$ R5 pairs, 5 lags, 3 days, both directions — 36,750 tests — was run. With Bonferroni correction at $\alpha = 0.05/36,750 = 1.36 \times 10^{-6}$: **zero pairs survive**. The empirical p -value distribution (Figure 2) matches the uniform-under-null prediction — 1,858 tests with $p < 0.05$ vs 1,838 expected by chance. *This is a confirmed null*, not a failure to find. At this tick resolution all R5 products move effectively simultaneously; contemporaneous correlations reflect joint price discovery rather than delayed propagation.

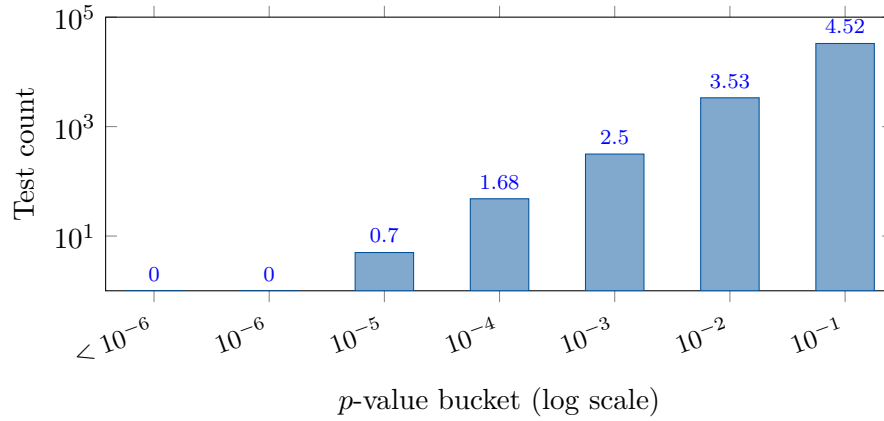


Figure 2: p -value distribution of the 36,750-test lead-lag CCF battery. Counts per bucket: 0, 0, 5, 48, 315, 3,361, 33,020 (left to right). The two leftmost buckets ($p < \alpha_{\text{Bonf}} = 1.36 \times 10^{-6}$) contain zero tests; the visible bars at those positions are stubs at count = 1 included only so all seven x-tick labels render under the log axis.

v42 — The Final Submitted Strategy

The decision moment. This is the single decision of the competition I will remember. v36 had won the local backtester at \$78,799. v42 came in at \$72,000 — a real \$6,799 behind. Every fibre of the “ship the winner” instinct said v36. But the Prosperity backtester runs only 10% of one day; the competition scores the full next day; and v36 had hit its number by blacklisting two products whose drift was negative in the early window and positive over the full day. Submitting v36 would have been chasing a number that was structurally not the objective. I shipped v42. I was genuinely terrified at the moment of submission — the sensation of overfitting becoming an instinct rather than an abstract concept — and I went to bed unsure I’d done the right thing. Three days later the official score came back: v42 algo + Ashflow manual = \$153,126 on Round 5, taking me from #592 to #346.

Lineage. The actually-traversed Round-5 version chain ran $v1 \rightarrow \dots \rightarrow v34 \rightarrow v35 \rightarrow v36 \rightarrow v37 \rightarrow v40 \rightarrow v41 \rightarrow \mathbf{v42}$. The two strategically load-bearing detours: **v36** (\$78,799 on the backtester) was the backtester champion, dropped on submission because it over-fit to the first-10% window by blacklisting two full-day-positive products. **v40** (not submitted) was the empirical demonstration of Insight 2: removing TIER3 (LIMIT 5 $\rightarrow$ 10 on 9 products) roughly doubled the adverse-selection loss, validating the linear-volume-cost model. v41 restored TIER3 plus added SNACKPACK_STRAWBERRY +10 from a HIGH-confidence sign-stable full-day-drift gate. v42 added the 8-product MM blacklist on top (the 7 ex-TIER3 products plus SLEEP_POD_LAMB_WOOL).

Construction. v42 differs from the v34 directional baseline by exactly two adds: the 8-product MM blacklist and SNACKPACK_STRAWBERRY +10. Concretely:

- 13 **directional positions** held at fixed ± 10 from the first tick of Day 5 (the multi-day-drift products).
- 11-**product MM blacklist** (LIMIT = 0): 3 zero-fill products inherited from v34 plus the 8 confirmed cross-version losers.
- 2 **small-loss TIER3 products** at LIMIT = 5 (MICROCHIP_RECTANGLE, OXYGEN_SHAKE_EVENING).
- **HEDGED_NO_SKEW pair**: SNACKPACK CHOC+VAN at structural $\rho = -0.916$, with bigger inner size (8 vs default 6) and no per-product inventory skew.
- Layer-1 directional runs first; Layer-2 per-product-tiered MM with skew fills the remaining capacity.

Realised PnL decomposition (from performance/algorithmic trading/round 5/581865.json, parsed post-result by re-applying v42’s strategy lists):

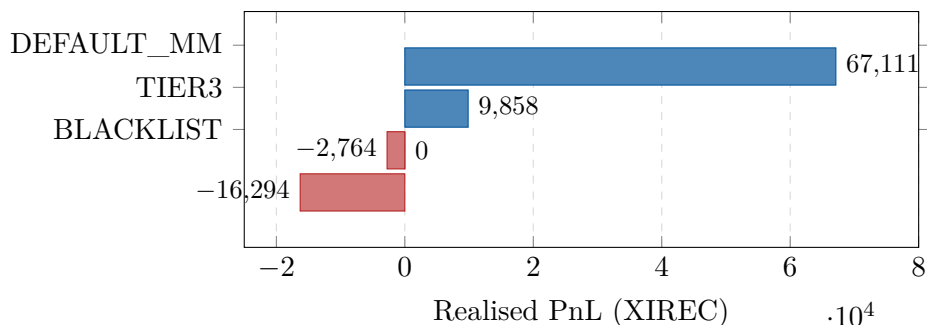


Figure 3: v42 Round-5 algo PnL by strategy bucket; sums to +\$57,911 matching the official Prosperity per-product algo JSON exactly.

The Prosperity backtester runs only the first 10% of Day 4, so under naive linear scaling a full-day score of $10 \times \$72,000 = \$720,000$ would have been the expected mark. The realised \$57,911 on Day 5 is $\sim 0.08\times$ that naive expectation — Day 5 was structurally weaker than the backtester window by a factor of roughly 12. The backtester is not a noisy estimator of the full-day score; it is a small-sample sub-window whose dynamics need not resemble the full day's. That is the operational meaning of “backtester $\neq$ competition”. Within the absolute scale that Day 5 actually paid, v42 did approximately as well as the strategy-design space allowed; the largest single ex-post miss was *not* a directional bet but MICROCHIP_SQUARE in DEFAULT_MM ($-\$18,636$), an adverse-selection product whose cross-version log history was inconsistent so it failed the blacklist criterion.

Lessons Learned

The 17 days changed how I think about three things: experiments, my own confidence, and the path forward. I'll take each in turn, in my own voice rather than the formal one the rest of this extract uses.

Round 3 nearly broke me, then taught me the most.

The lowest moment of the competition was the first official Round-3 submission. I had been at a steady $\sim$ \$10K PnL through Round 2; the first Round-3 submission scored close to \$0. The introduction of options — Black–Scholes, implied volatility, the Greeks, delta hedging — on top of an already-failing strategy felt like falling into a well with no end. I genuinely thought, briefly, that I was finished.

Three days later, the same Round-3 algorithm earned +\$40K on the official computation day, and my rank jumped from $\sim$ #1,500 to #800. The oscillation taught me more than any of the wins. *Local backtest numbers can be wildly off the real engine. Emotional response to a bad number tells me nothing about the underlying signal. Persistence through a bad rank is just the price of being there at the next reading.* I went into Round 4 knowing this. I came out of Round 5 still knowing it.

The backtester taught me what overfitting actually feels like.

The single most consequential decision I made was choosing to submit v42 over v36 in Round 5. v36 scored \$78,799 on the Prosperity backtester; v42 scored \$72,000. A real, measurable \$6,799 regression. v36 was the local optimum and submitting it was the obvious move. But the Prosperity backtester runs only the first 10% of Day 4, while the competition scores the full Day 5. v36 hit that 10%-window optimum by dropping two products whose drift was negative in the early window but positive over the full day. Submitting v36 would have been chasing a number that was structurally not the objective.

I had read about overfitting in every ML textbook. This was the first time I felt what it actually does to your decisions in real time: the pull towards the higher visible score, the discomfort of shipping the regression, the rationalisation that maybe my full-day estimate is wrong. It terrified me to ship the lower number. It also made the difference between “overfitting” as a textbook concept and overfitting as a working framework I now know how to detect on my own work.

Manuals carried 59.4% of the cumulative score — and that says something.

I put far more hours into the algorithmic side than the manuals. The manuals delivered \$227,968 of the \$383,727 final cumulative total. My algorithmic rank (#537) was meaningfully weaker than my manual rank (#204). I would like to claim this as a lesson about asymmetric-information manuals being where the alpha actually lives, and that's true — but it is also a lesson about effort allocation. The manuals were essentially closed-form game-theory puzzles: read the rules carefully, write the right code once. The algorithm rewards continuous iteration and patient calibration. Both are skills; I now know which I need to deepen for the role I want.

The before-and-after.

Before Prosperity 4, I had no quant project on my CV, no track record in this field, and no concrete picture of how a serious quant works. Seventeen days later I have a framework. I know what to study, what to build next, and how to be patient with the gap between where I am and where I want to be. The result — top 1.84%, country #11, monotone improvement across all five rounds from $\sim$ #2,000 to #346 — is not the end-state I want, but it is a concrete signal that the trajectory works if I keep at it. I'm proud of it. I am also writing this report partly for the version

of myself that did not yet know the trajectory worked, and partly for the friend or recruiter who, like me at the start, knows almost nothing about this competition.

Day-1 Plan for Prosperity 5

If I entered next year's competition tomorrow, four concrete actions would change from how I started this one. I am writing them down here because I will forget by next April otherwise.

1. Manuals first, every round, before the algorithm. This year I left the manuals to the last few hours of each round, after the algorithmic work was already deep. The manuals are closed-form: their PnL ceiling is fixed once the puzzle is decoded, and the optimisation is bounded. The algorithm is open-ended; it will absorb every minute you give it. Doing manuals first lets the algorithm time become the residual — not the other way round. This alone would have changed my Round-3 Bio-Pods and Round-5 Ashflow submissions, both of which I did under time pressure hours before the deadline.

2. Understand the signal before implementing the strategy. I caught myself multiple times this year writing code first and rationalising the alpha after. Round 5 made the cost of this concrete: the v40 TIER3-removal mistake came from not actually understanding why TIER3 was set the way it was in v34 to begin with. Next year, every strategy goes through a written one-paragraph justification (*what is the structural source of edge here*) before any implementation.

3. Use the LLM as a research assistant, not as a code-writer-of-record. I am going to be precise about this because it was a real failure mode. When I was tired or stuck, I asked Claude to write a function and shipped it without fully reading what it did. Some of the bugs that cost me the most — including the R3 website-backtester 0-PnL incident — happened in code I had not actually read line-by-line. Next year: the LLM is a research collaborator for understanding edges, exploring concepts, and explaining unfamiliar mathematics. It is not a code generator I deploy without review. Every line of trader code I submit, I read.

4. Read the code I write. This is a corollary of (3) but deserves its own line because I did not always do it. Before any submission, read every line of the trader file top to bottom, with intent. This sounds basic. It is.

What the competition is actually testing.

In retrospect, IMC Prosperity is not a quantitative-finance test — it is a *methodology test*. The mathematics is undergraduate-tractable (Black–Scholes with $r = 0$, AR(1), Bayesian-Nash, Granger). The hard part is the discipline: pre-register the out-of-sample day, calibrate to the right window, take null results seriously, refuse to optimise to easy-to-measure but not-actually-the-objective scores. The teams that finished above me likely did not have better mathematics. They had better experimental hygiene. That is what I have in front of me to learn.

What's next

This report sits at the start of a longer trajectory, not the end of a project. Concretely, I am: (1) preparing for quantitative-research / quantitative-trading interview pipelines at IMC and the firms I met in Zurich — the curriculum is probability, statistics, market microstructure, and stochastic calculus, in roughly that order; (2) building a second research project — a small-budget systematic strategy on liquid retail-broker data — to take the experimental discipline I learned here and apply it to real markets where the overfitting cost is not synthetic XIREC; (3) entering Prosperity 5 next year, with the Day-1 Plan above written down and posted somewhere I will see it on day 0. If you are a recruiter or admissions reader and want to talk about any of this, please reach out — the rest of the trajectory is what I am building toward.

If You Want to Go Deeper

This extract is a ~ 11 -page distillation. The substance behind every claim is in the full $\sim 11,500$ -line technical report at `report.tex`. Pointers, by interest:

For the full Round-by-Round narrative. `report.tex` Part III (“Round-by-Round Analysis”) — one chapter per round (R0 tutorial, R1, R2, R3 options, R4 counterparty, R5 50-product), each with EDA, hypothesis, method, result, and lesson.

For the deep R5 phase pipeline. Part III § “Round 5: Building Directional Positions (Phase 13)” through “Round 5: Passive Market-Making Doubles the Backtester PnL (Phase 17)”. This is where the $v34 \rightarrow v42$ strategy evolution lives, including $v40$ ’s TIER3-removal disaster as the controlled empirical demonstration of Insight 2.

For the manual game-theory derivations. `report.tex` § “Manual Trading: Intarian Welcome Auction” (R1), § “Part B: MAF bid game theory” (R2), § “Part C: Manual ‘Invest & Expand’ optimization” (R2), § “Bio-Pod Manual Auction: Bayesian-Nash Solution” (R3), § “Round 4 Manual: Vanilla Just Isn’t Exotic Enough” (R4 AETHER_CRYSTAL exotic options), § “Manual Challenge — Ashflow Alpha (Ignith Exchange)” (R5).

For the lead-lag null result. `report.tex` § “Round 5: Lead-Lag Analysis: Structural Sequence Signals”. Includes the formal Granger framework, Hasbrouck’s information share, and the full 36,750-test CCF battery.

For the $v42$ final-design. `report.tex` § ?? “ $v42$ — The Final Submitted Strategy” (in the full document).

For the Prosperity Vault second-brain. `prosperity-vault/` contains ~ 115 Obsidian pages organised into Concepts, Strategies, Products, Rounds, Backtests, Parameters, and Performance — the underlying knowledge graph for everything in the report.

Canonical references cited in the body

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